

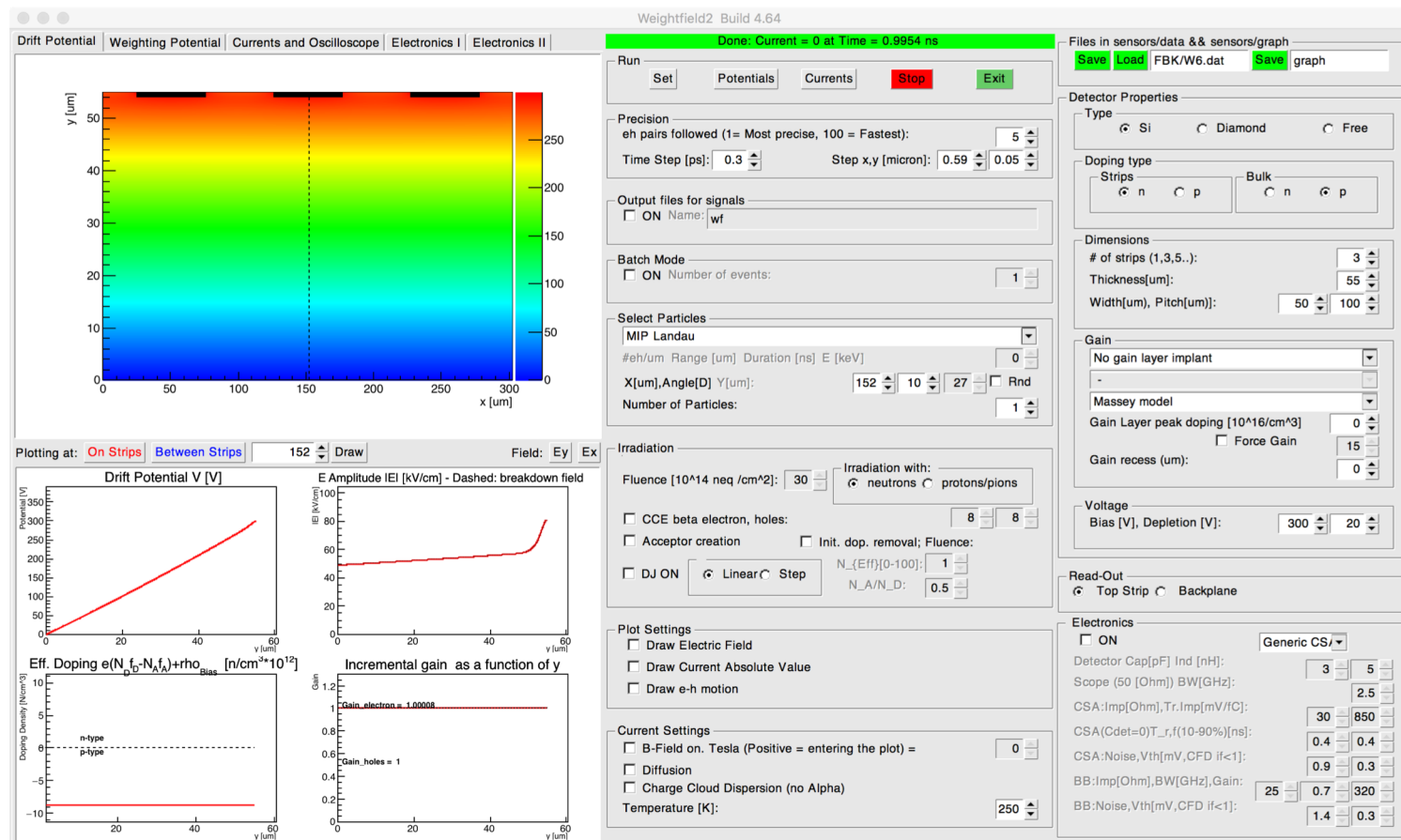
Weightfield2

Available at:

<http://personalpages.to.infn.it/~cartiglia/Weightfield2/Main.html>

It requires Root build from source, it is for Linux and Mac.

It will not replace TCAD, but it helps in understanding the sensors response

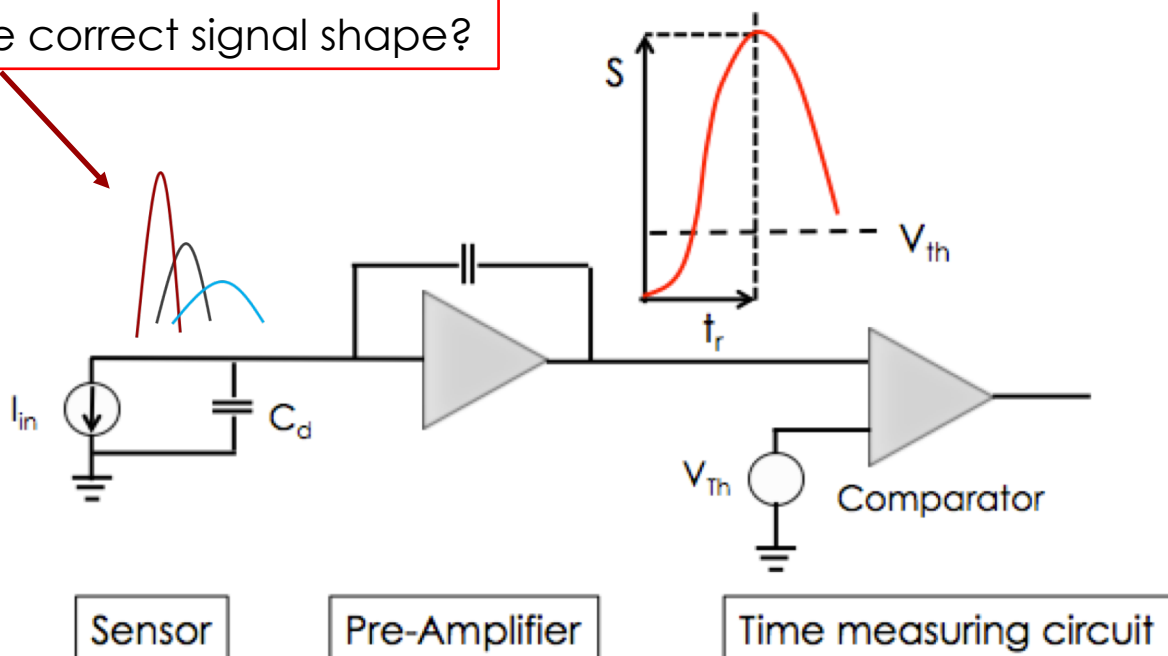


Why did we design WF2?

From Angelo's keynote talk:

- Electronics for ps timing is (in principle) already there!
- **But what about sensors?**
- Sensor and front-end codesign essential to achieve best possible timing

What is the correct signal shape?



Weightfield2

Highlights:

- It is completely open source
- it's fast
- It generates the signal from several sources (MIP, alpha, lasers..)
- Runs in batch mode writing output files
- It loads/save configurations
- It has basics electronics simulation

It crashes occasionally

How to use it:

Obtain the last version from

<http://personalpages.to.infn.it/~cartigli/Weightfield2/Main.html>

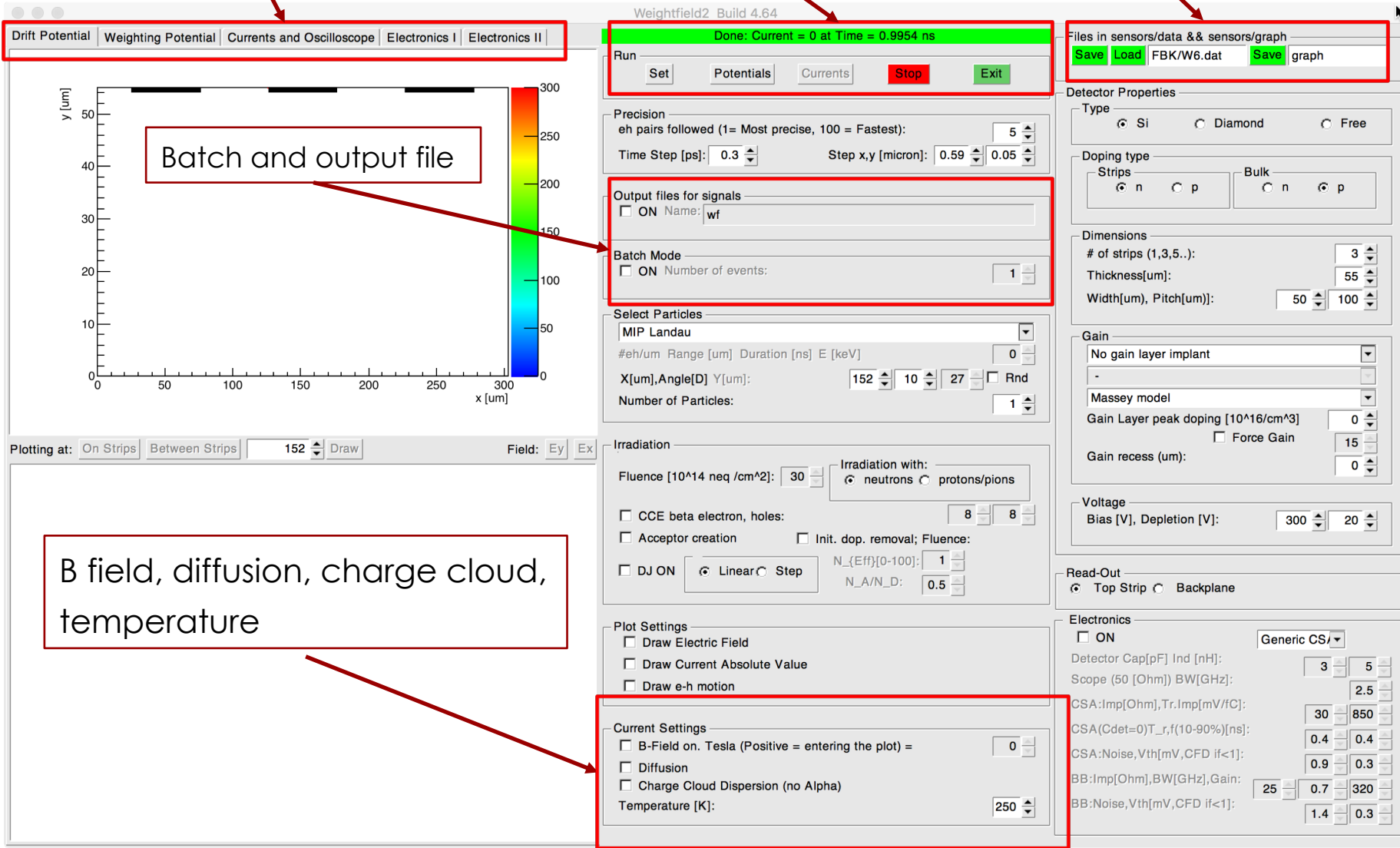
- 1) From the download page, get the latest version
- 2) Unzip it and then type:
- 3) Make or 3-bis) make -f Makefile_MacOS10.10_root6
- 4) ./weightfield

WF2 layout

4 tabs: field, Weighting field, currents, and electronics

Controls

The program can save/load your configuration and the plots



Batch and output file

Controls

The program can save/load your configuration and the plots

B field, diffusion, charge cloud, temperature

Step 1: select your sensor

Weightfield2 Build 4.64

Done: Current = 0 at Time = 0.9954 ns

Run: Set Potentials Currents Stop Exit

Precision: eh pairs followed (1= Most precise, 100 = Fastest): 5
Time Step [ps]: 0.3 Step x,y [micron]: 0.59 0.05

Output files for signals: ☐ ON Name: wf

Batch Mode: ☐ ON Number of events: 1

Select Particles: MIP Landau
#eh/um Range [um] Duration [ns] E [keV]: 0
X[um],Angle[D] Y[um]: 152 10 27 ☐ Rnd
Number of Particles: 1

Irradiation: Fluence [10¹⁴ neq /cm²]: 30
Irradiation with: ☒ neutrons ☐ protons/pions
☐ CCE beta electron, holes: 8 8
☐ Acceptor creation ☐ Init. dop. removal; Fluence:
☐ DJ ON ☒ Linear ☐ Step N_{Eff}[0-100]: 1
N_A/N_D: 0.5

Plot Settings: ☐ Draw Electric Field
☐ Draw Current Absolute Value
☐ Draw e-h motion

Current Settings: ☐ B-Field on. Tesla (Positive = entering the plot) = 0
☐ Diffusion
☐ Charge Cloud Dispersion (no Alpha)
Temperature [K]: 250

Files in sensors/data && sensors/graph: Save Load FBK/W6.dat Save graph

Detector Properties:

- Type: ☒ Si ☐ Diamond ☐ Free
- Doping type: Strips ☒ n ☐ p Bulk ☐ n ☒ p
- Dimensions: # of strips (1,3,5..): 3 Thickness[um]: 55 Width[um], Pitch[um]: 50 100
- Gain: No gain layer implant - Massey model Gain Layer peak doping [10¹⁶/cm³]: 0 ☐ Force Gain 15 Gain recess (um): 0
- Voltage: Bias [V], Depletion [V]: 300 20

Read-Out: ☒ Top Strip ☐ Backplane

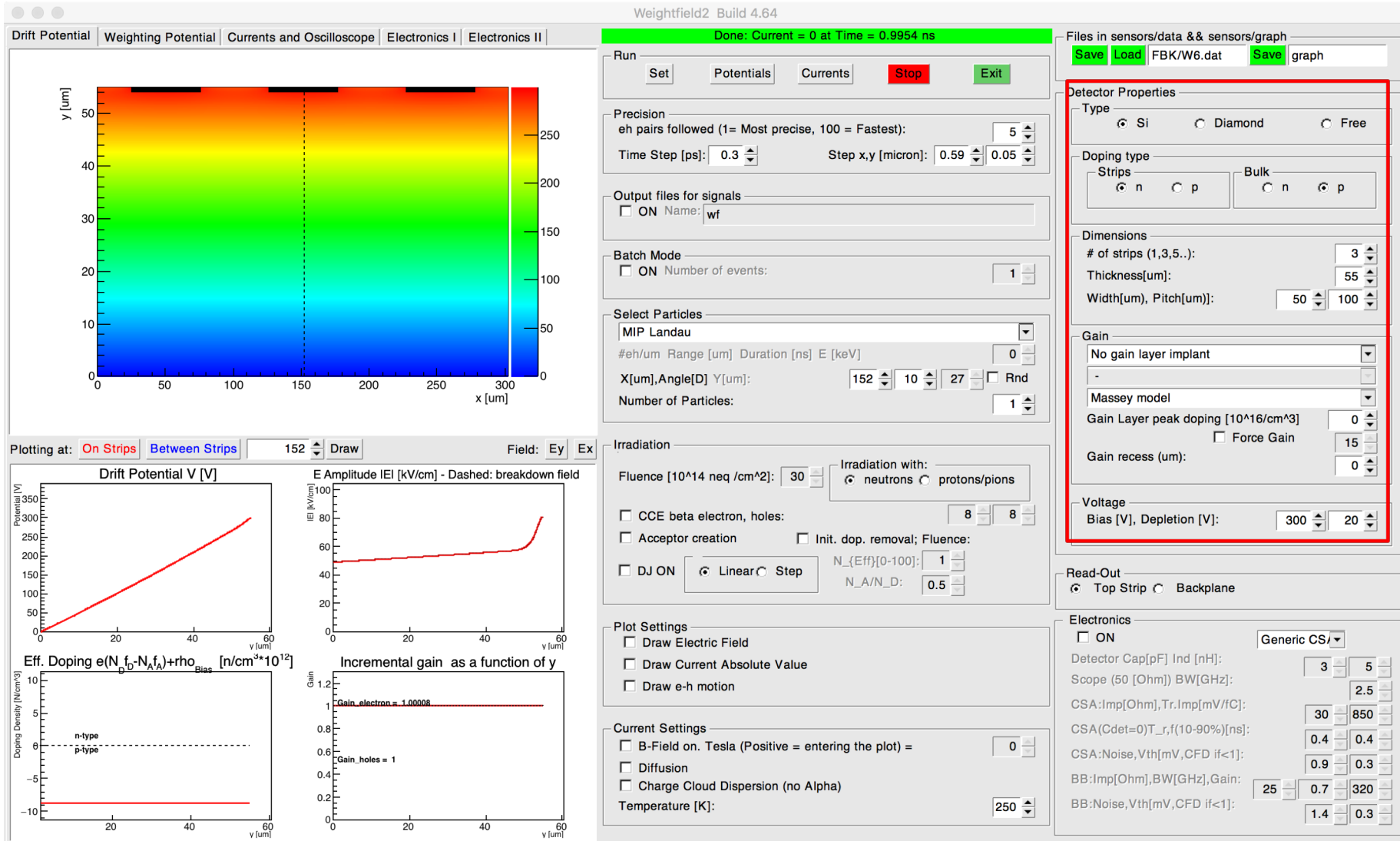
Electronics: ☐ ON Generic CS
Detector Cap[pF] Ind [nH]: 3 5
Scope (50 [Ohm]) BW[GHz]: 2.5
CSA:Imp[Ohm],Tr.Imp[mV/fC]: 30 850
CSA(Cdet=0)T_r,f(10-90%)[ns]: 0.4 0.4
CSA:Noise,Vth[mV,CFD if<1]: 0.9 0.3
BB:Imp[Ohm],BW[GHz],Gain: 25 0.7 320
BB:Noise,Vth[mV,CFD if<1]: 1.4 0.3

Plotting at: On Strips Between Strips 152 Draw Field: Ey Ex

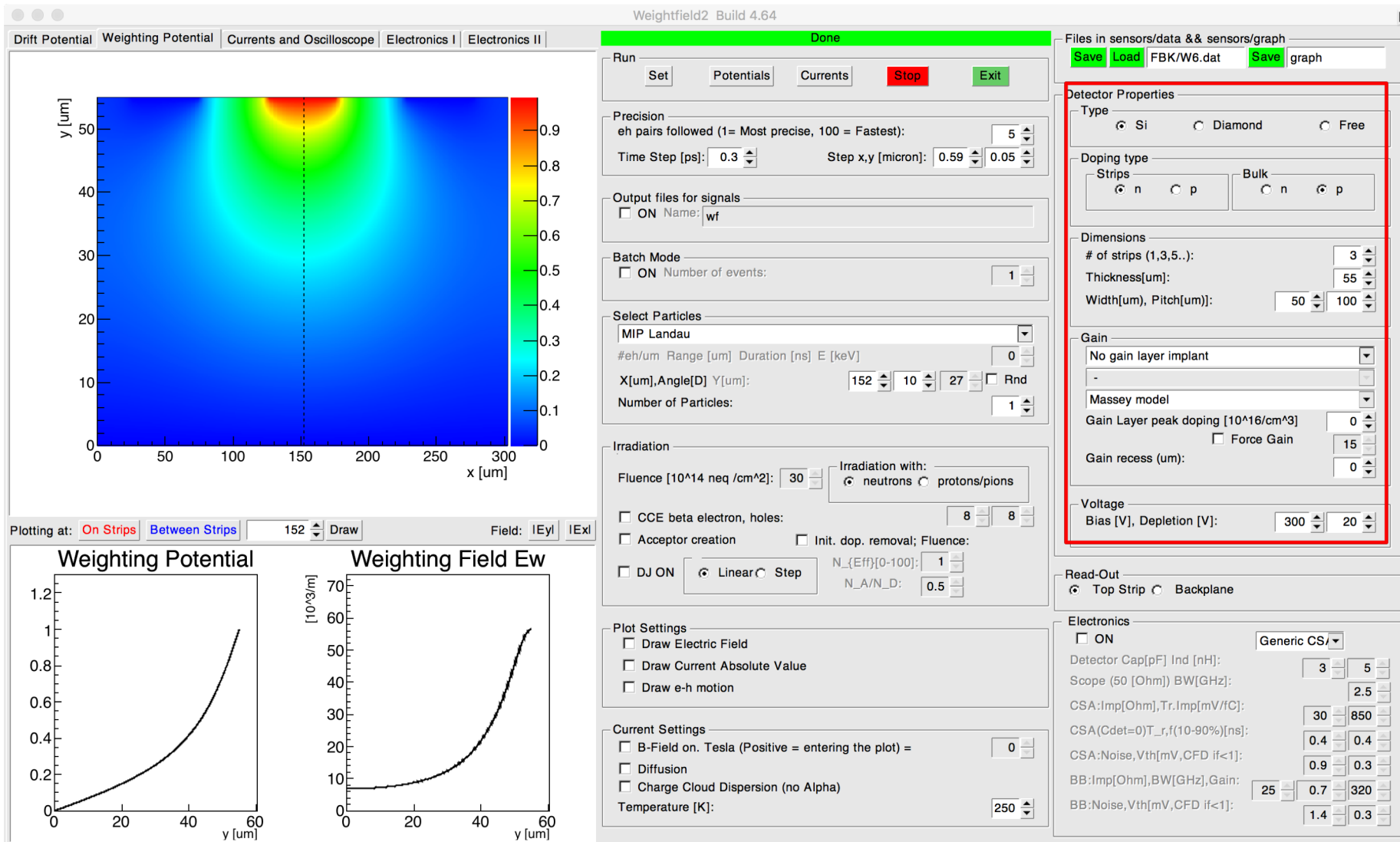
Fields: under the hood

- The program loads your geometry
- Compute the silicon resistivity from the depletion voltage
- It uses an iterative method to compute:
 - The electric field
 - The weighting field

Step 1: E field

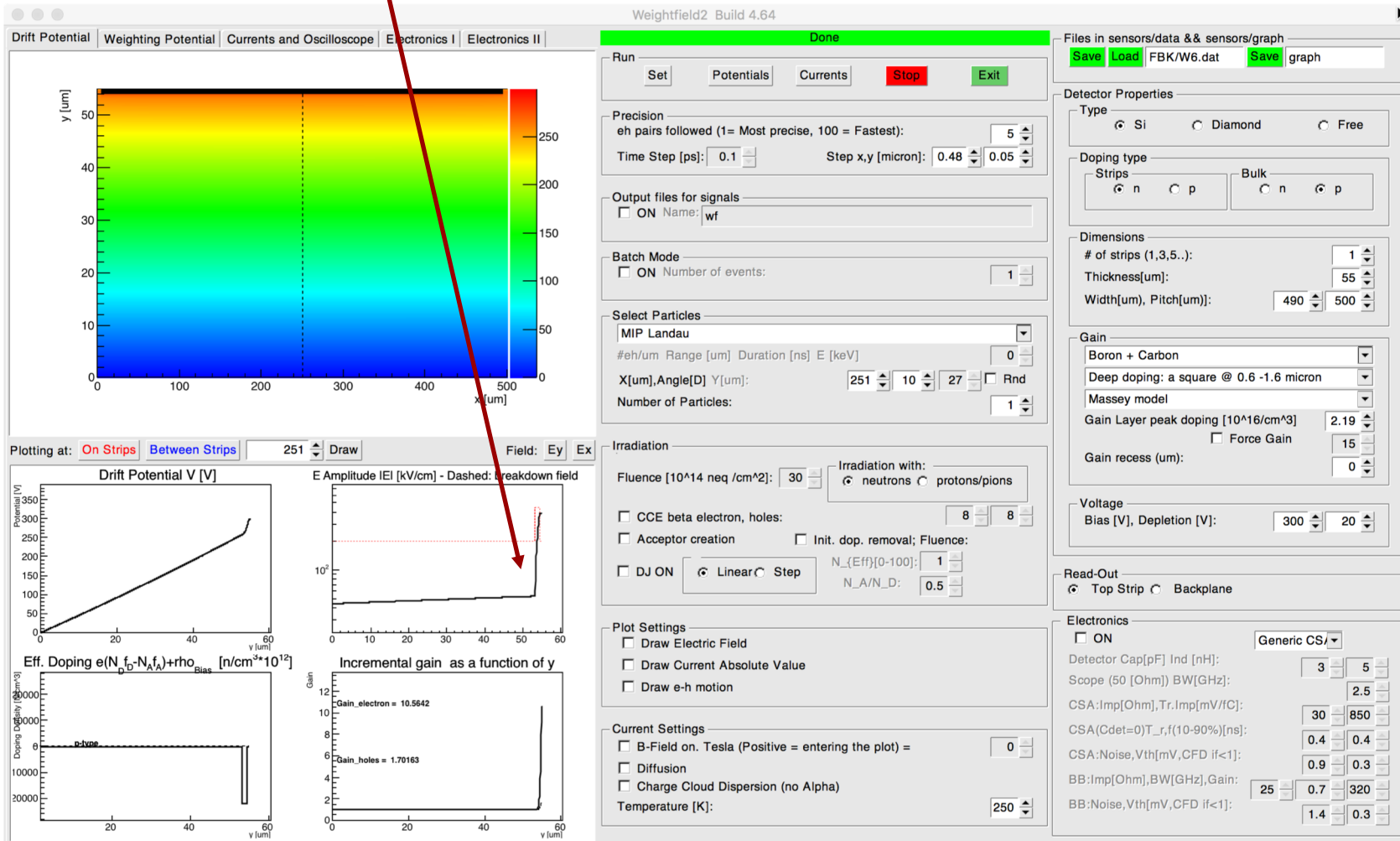


Step 1: W field



Select your sensor: does it have gain?

- The program implements a gain layer
- It computes the contribution from the additional doping to the electric field

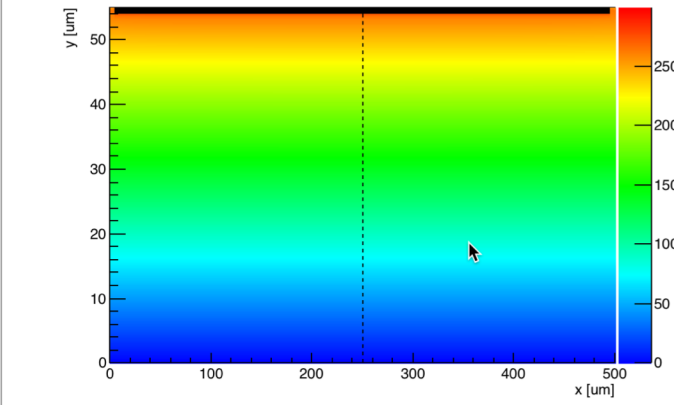


Step 2: select the particle

Grab File Edit Capture Window Help

Weightfield2 Build 4.64

Drift Potential Weighting Potential Currents and Oscilloscope Electronics I Electronics II



Done

Run Set Potentials Currents Stop Exit

Precision
eh pairs followed (1= Most precise, 100 = Fastest): 5

Time Step [ps]: 0.1 Step x,y [micron]: 0.48 0.05

Output files for signals
☐ ON Name: wf

Batch Mode
☐ ON Number of events: 1

Select Particles

- MIP Landau
- MIP: uniform Q, Qtot = $q \cdot [\#eh/um] \cdot Height$
- MIP: NON uniform Q, Qtot = $q \cdot [\#eh/um] \cdot Height$
- MIP Landau
- Laser (1064 nm): Top-TCT, Q = $q \cdot [\#eh/um] \cdot Height$
- Laser (1064 nm): Edge-TCT, Q = $q \cdot [\#eh/um] \cdot Height$
- Edge MIP Landau

Irradiation

Fluence [10^{14} neq /cm²]: 30 Irradiation with: ☒ neutrons ☐ protons/pions

☐ CCE beta electron, holes: 8 8

☐ Acceptor creation ☐ Init. dop. removal; Fluence:

☐ DJ ON ☒ Linear ☐ Step

$N_{eff}[0-100]: 1$

$N_A/N_D: 0.5$

Plot Settings

☐ Draw Electric Field

☐ Draw Current Absolute Value

☐ Draw e-h motion

Current Settings

☐ B-Field on. Tesla (Positive = entering the plot) = 0

☐ Diffusion

☐ Charge Cloud Dispersion (no Alpha)

Temperature [K]: 250

Files in sensors/data && sensors/graph

Save Load FBK/W6.dat Save graph

Detector Properties

Type ☒ Si ☐ Diamond ☐ Free

Doping type

Strips ☒ n ☐ p Bulk ☐ n ☒ p

Dimensions

of strips (1,3,5..): 1

Thickness[um]: 55

Width[um], Pitch[um]: 490 500

Gain

Boron + Carbon

Deep doping: a square @ 0.6 -1.6 micron

Massey model

Gain Layer peak doping [$10^{16}/cm^3$]: 2.19

☐ Force Gain 15

Gain recess (um): 0

Voltage

Bias [V], Depletion [V]: 300 20

Read-Out

☒ Top Strip ☐ C ☐ Backplane

Electronics

☐ ON Generic CS

Detector Cap[pF] Ind [nH]: 3 5

Scope (50 [Ohm]) BW[GHz]: 2.5

CSA:Imp[Ohm], Tr.Imp[mV/IC]: 30 850

CSA(Cdet=0)T_r,f(10-90%)[ns]: 0.4 0.4

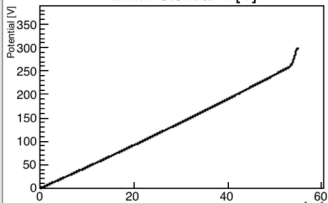
CSA:Noise,Vth[mV,CFD if<1]: 0.9 0.3

BB:Imp[Ohm], BW[GHz], Gain: 25 0.7 320

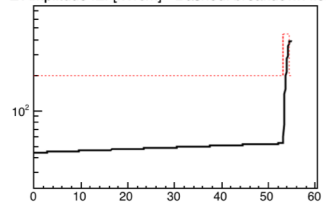
BB:Noise,Vth[mV,CFD if<1]: 1.4 0.3

Plotting at: On Strips Between Strips 251 Draw Field: Ey Ex

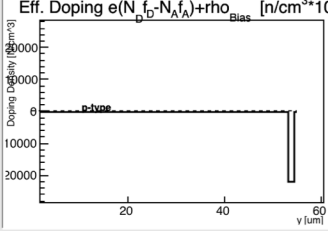
Drift Potential V [V]



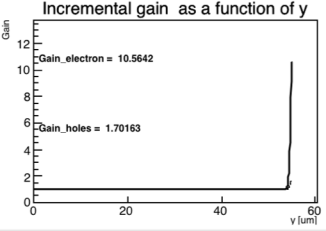
E Amplitude |E| [kV/cm] - Dashed: breakdown field



Eff. Doping $e(N_D - N_A) + \rho$ [n/cm³ * 10¹²]



Incremental gain as a function of v

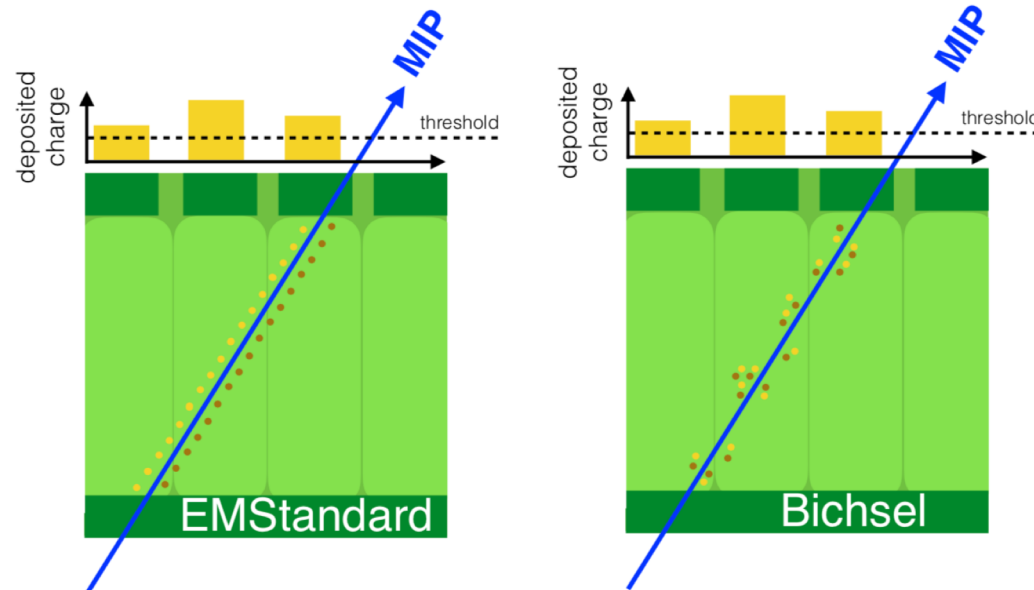


Gain_electron = 10.5642

Gain_holes = 1.70163

Landau: under the hood

The program uses GEANT4 with the photo-absorption ionization (PAI) model to generate non uniform charge depositions



Results cross-checked with several publications, for example:

The Impact of Incorporating Shell-corrections to Energy Loss in Silicon

Fuyue Wang, Dong Su, Benjamin Nachman, Maurice Garcia-Sciveres, and Qi Zeng
arXiv:1711.05465v2 [physics.ins-det]

“ The ionization energy loss fluctuation in very thin silicon sensors significantly deviates from the Landau distribution. Therefore, we have developed a charge deposition setup that implements the Bichsel straggling function, which accounts for shell-effects. ”

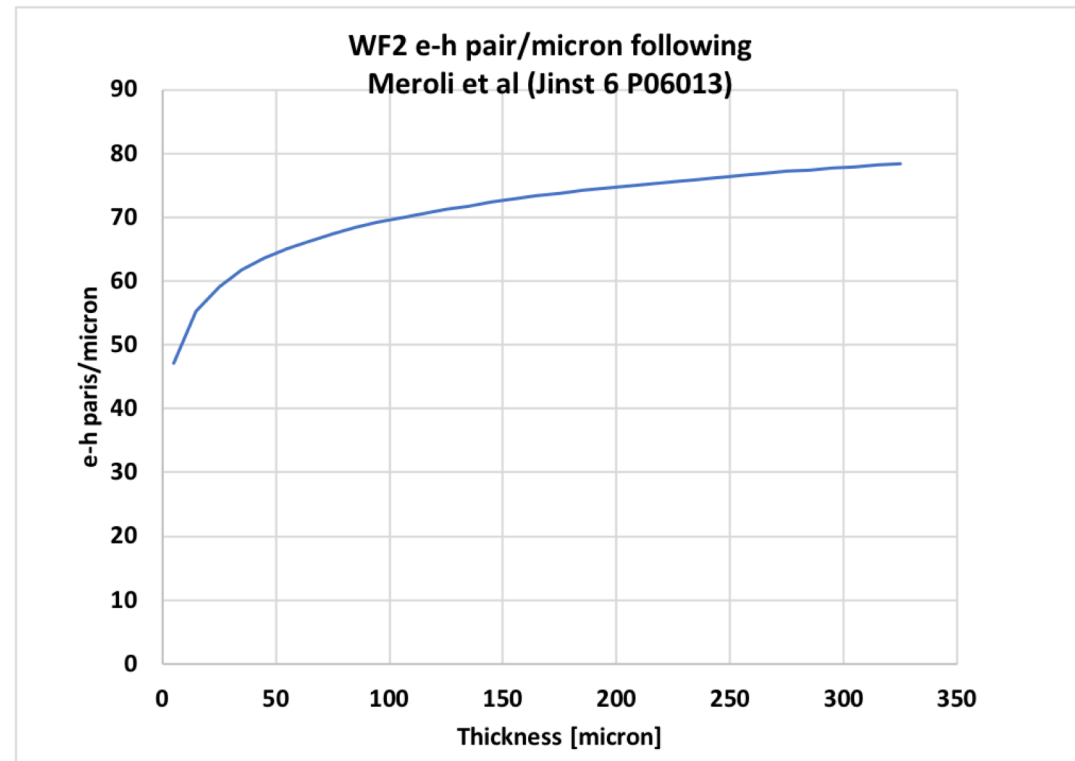
Landau: under the hood

Landau distribution

$$\text{MPV} = d * [0.027 * \ln(d) + 0.126]$$

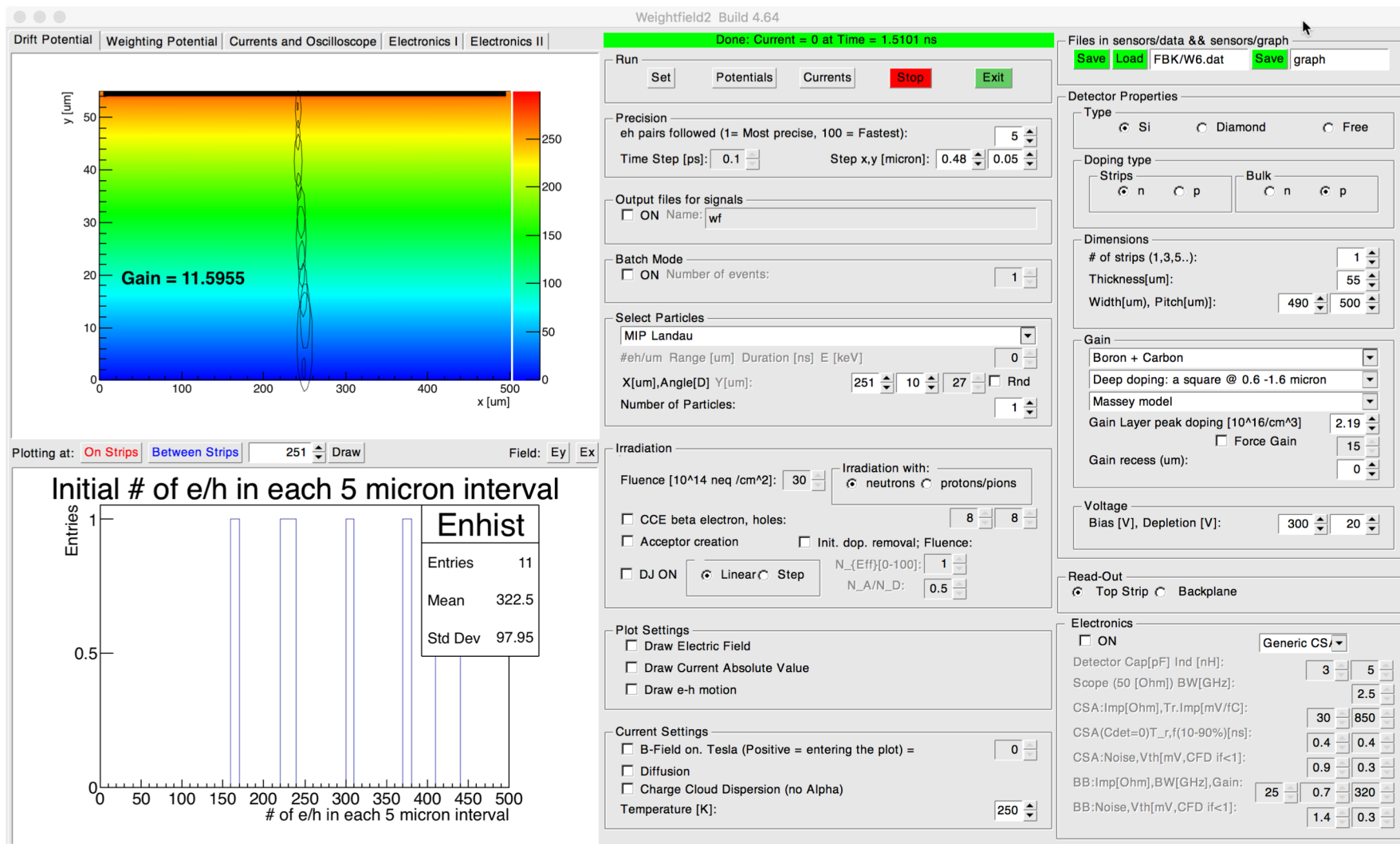
$$\text{FWHM} = 0.31 * d^{0.81}$$

$$\frac{\text{FWHM}}{\text{MPV}} = 2.1 * d^{-0.3}$$



Following Meroli et al (Jinst 6 P06013), these are the parameterizations of the MPV and FWHM as a function of the sensor thickness d for the Landau distribution in silicon

Step 3: charge carriers drift



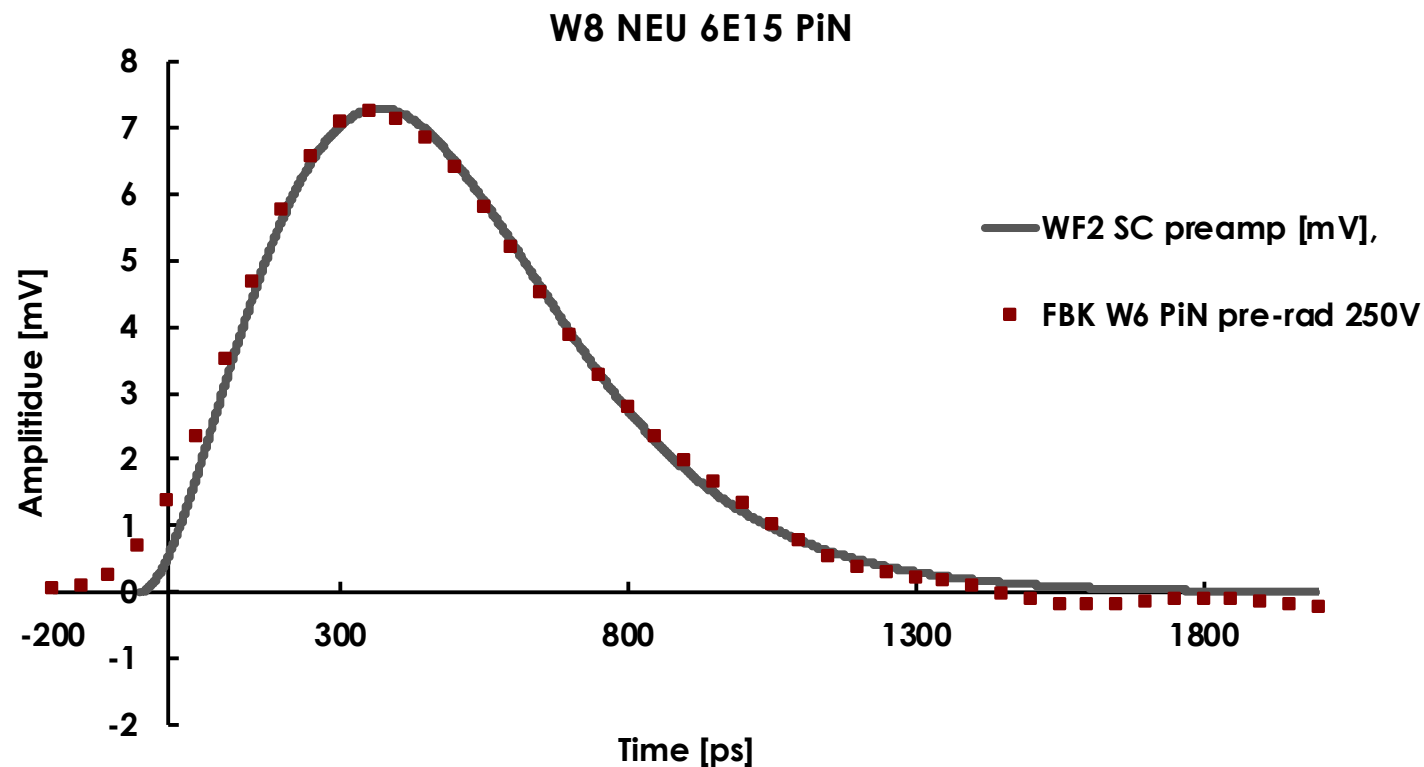
drift: under the hood

Current is generated using Ramo's theorem: $i(t) = qv(t)E_w$

$$I_{tot}(t_j) = \sum_{k=1}^n I_k(t_j) = -q \sum_{k=1}^n \overrightarrow{v_k(t_j, x_k)} \cdot \overrightarrow{E_w}(x_k)$$

	Electrons	Holes
$\mu(T) \left[m^2/Vs \right]$	$0.1414 \left(\frac{T}{300K} \right)^{-2.5}$	$0.0470 \left(\frac{T}{300K} \right)^{-2.2}$
$\beta(T)$	$1.09 \left(\frac{T}{300K} \right)^{0.66}$	$1.213 \left(\frac{T}{300K} \right)^{0.17}$
$v_{sat}(T) [m/s]$	$1.07e5 \left(\frac{300K}{T} \right)^{0.87}$	$8.35e4 \left(\frac{300K}{T} \right)^{0.52}$
$v(x, T) [m/s]$	$\frac{\mu_e(T)E_d(x)}{\sqrt[1/\beta_e(T)]{1 + \left(\frac{\mu_e(T)E_d(T)}{v_{e,sat}(T)} \right)^{\beta_e(T)}}}$	$\frac{\mu_h(T)E_d(x)}{\sqrt[1/\beta_h(T)]{1 + \left(\frac{\mu_h(T)E_d(x)}{v_{h,sat}(T)} \right)^{\beta_h(T)}}}$

WF2 – Data: current in PiN



gain: under the hood

If the electric field is high enough, carriers multiply

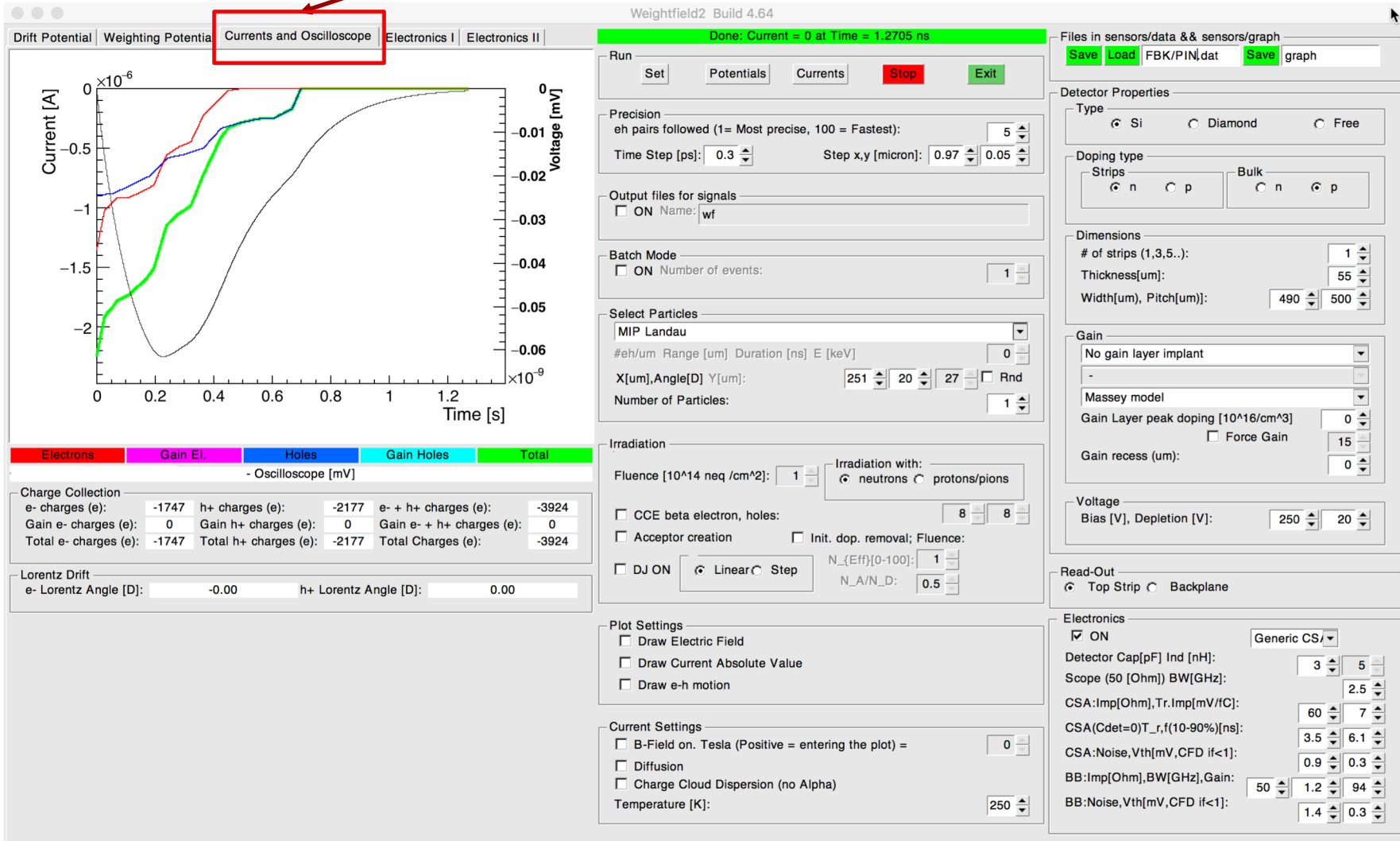
$$N_e(x) = N_e e^{\beta x}; \quad N_h(x) = N_h e^{\alpha x}$$

$$\alpha = A_n \exp\left\{-\frac{B_n}{E}\right\} ;$$
$$\beta = A_p \exp\left\{-\frac{B_p}{E}\right\} ,$$

$$B_{n,p}(T) = C_{n,p} + D_{n,p} T$$

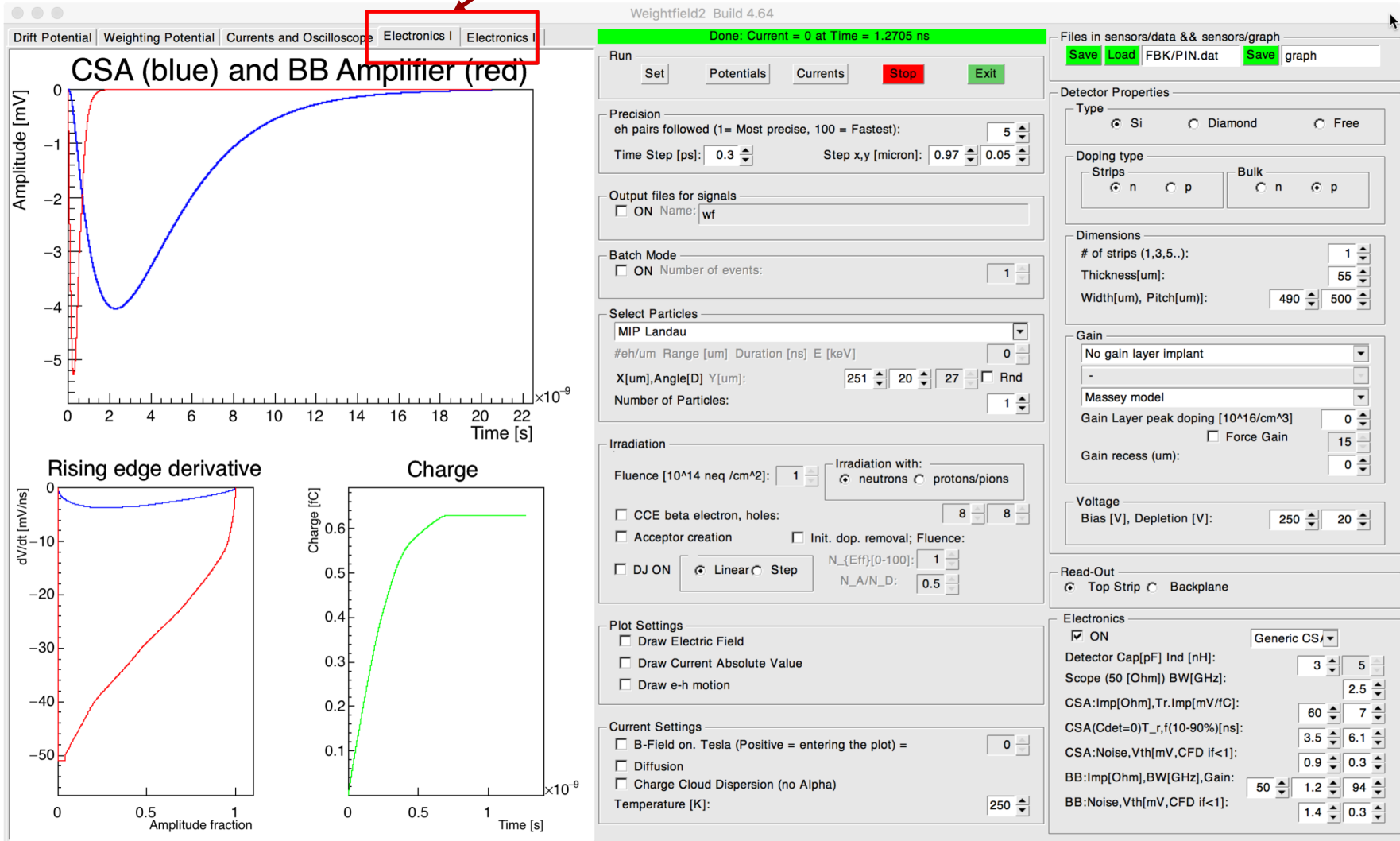
Currents

Current tabs

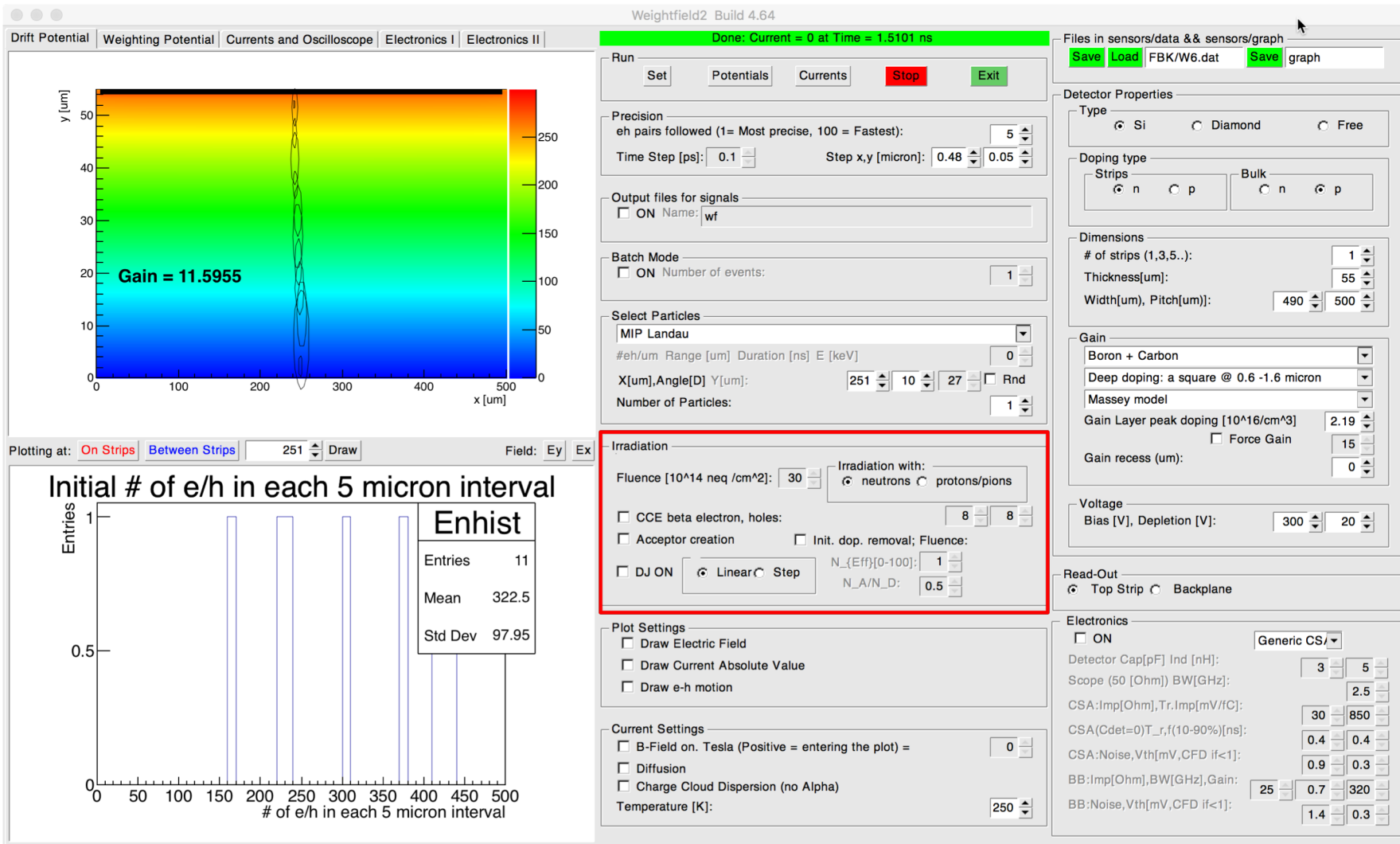


Electronics

Electronics tabs



Step 4: radiation damage



Step 4: under the hood

Charge trapping with fluence ϕ :

$$i(t) = i(t)_{new} e^{-t/\tau}$$

$$\tau = \beta \phi \leftarrow \text{model under discussion}$$

Acceptor removal:

$$N(\phi) = N(0) * e^{-c\phi}$$

Acceptor creation:

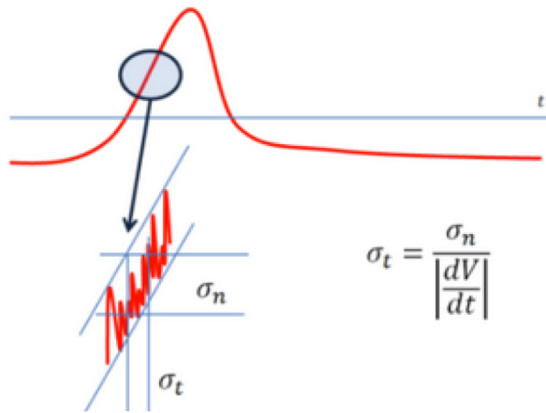
$$N(\phi) = \beta \phi$$

WF2: predictions

$$\sigma_t = \left(\frac{N}{dV/dt} \right)^2 + (\text{Landau Shape})^2 + \text{TDC}$$

Usual "Jitter" term

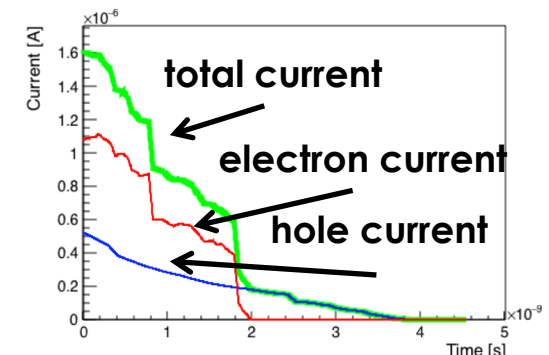
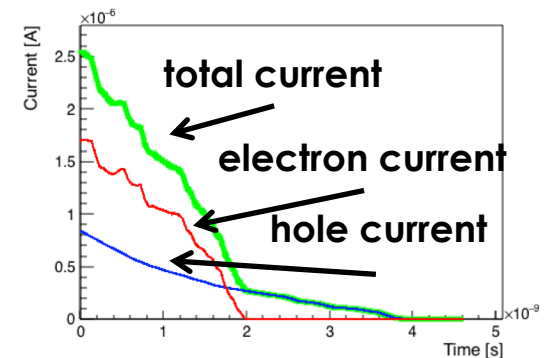
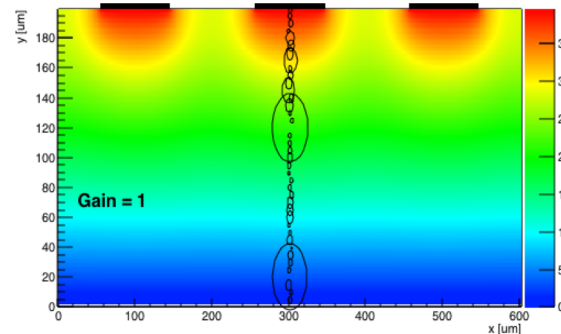
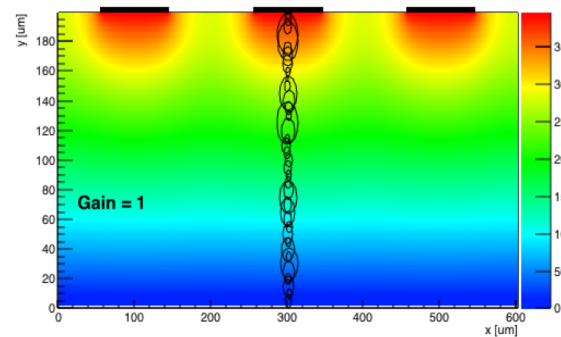
Here enters everything that is "Noise" and the steepness of the signal



Need large dV/dt

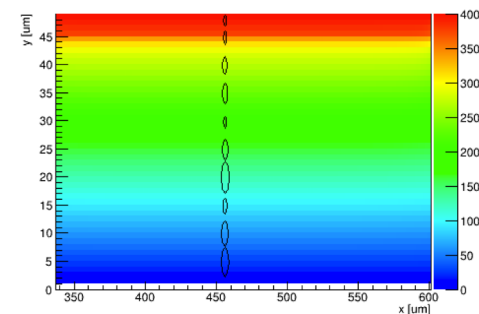
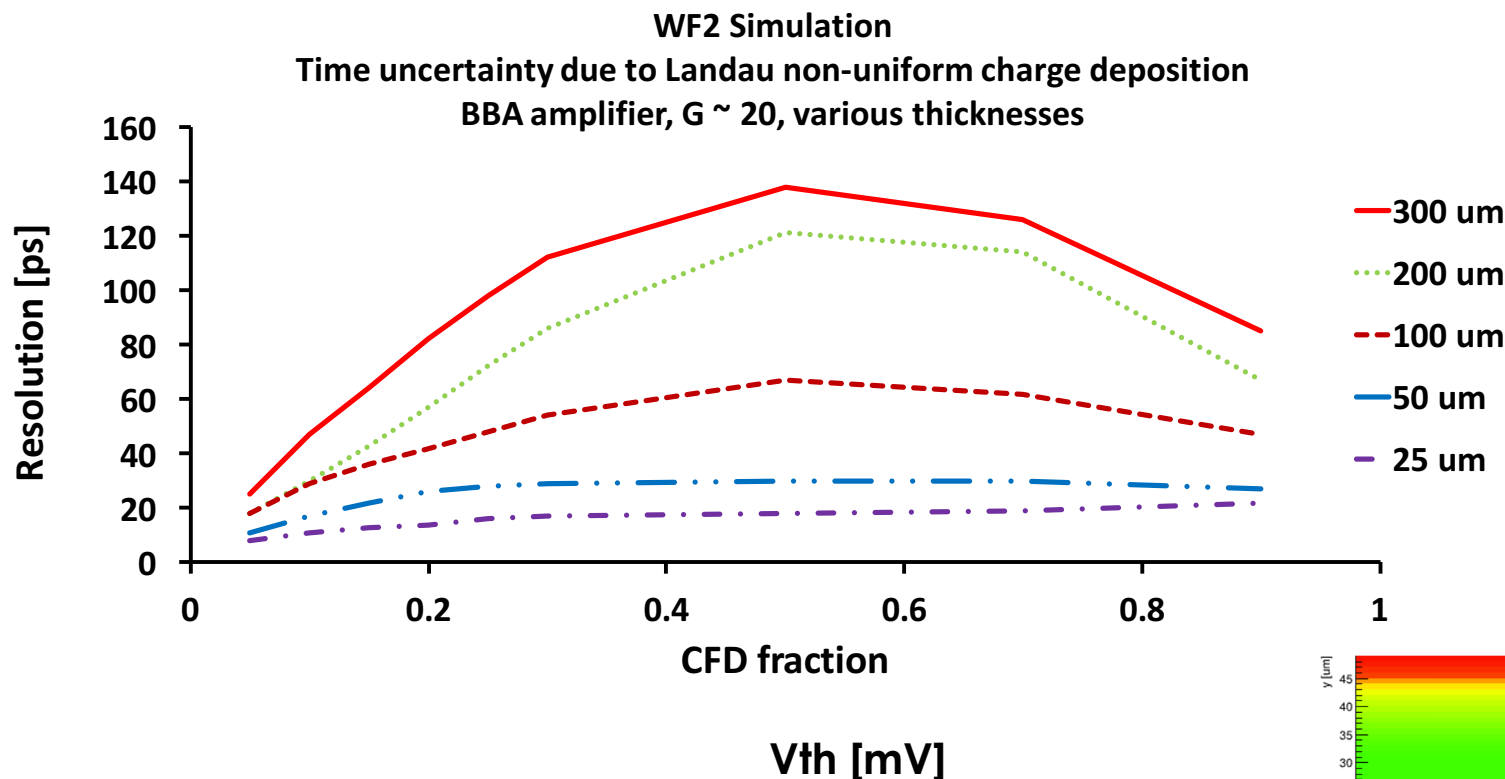
Time walk: Amplitude variation, corrected in electronics

Shape variations: non homogeneous energy



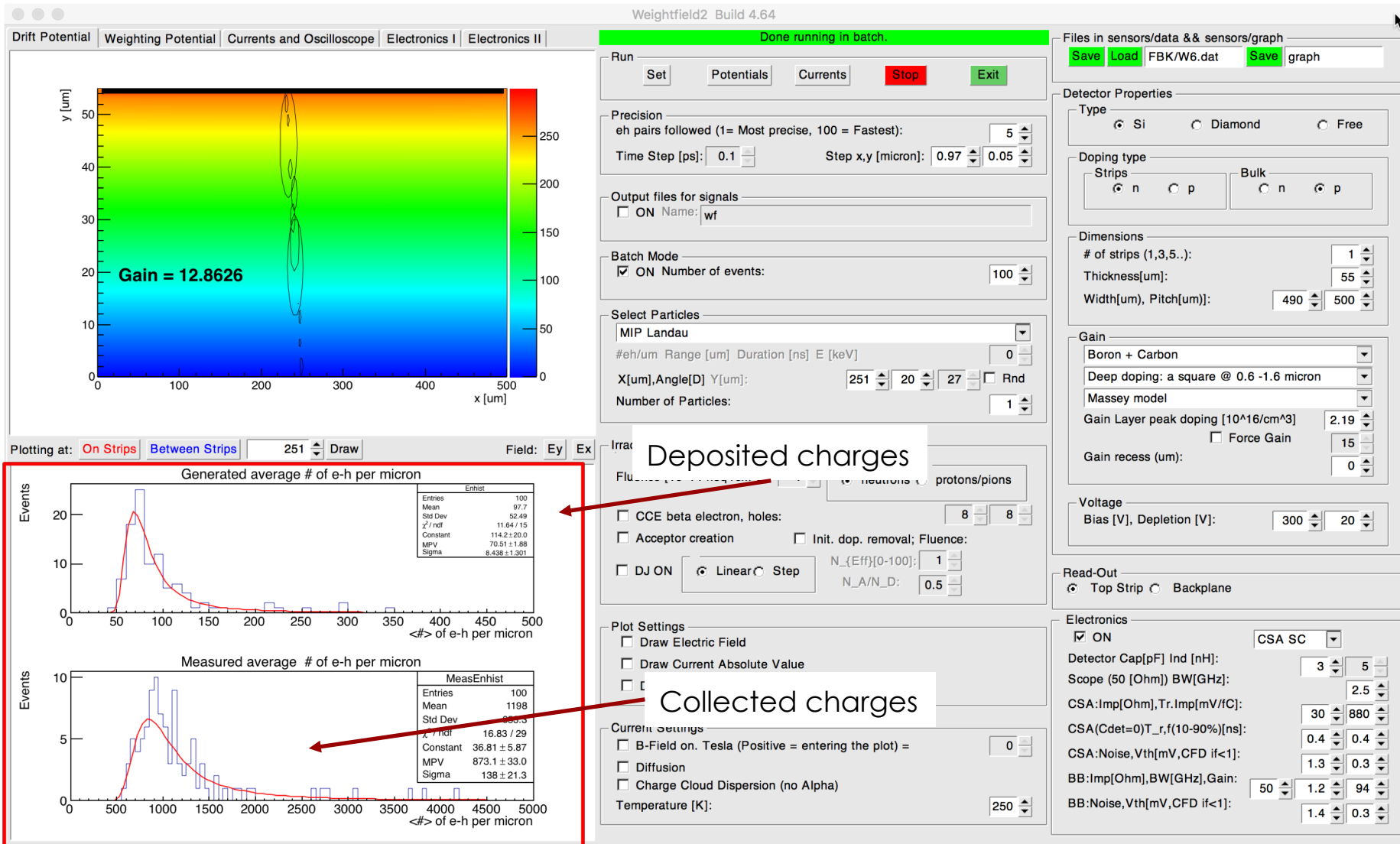
Non uniform charge deposition along the track

This is a physical limit to time resolution:
Need to use thin detectors and low comparator threshold.

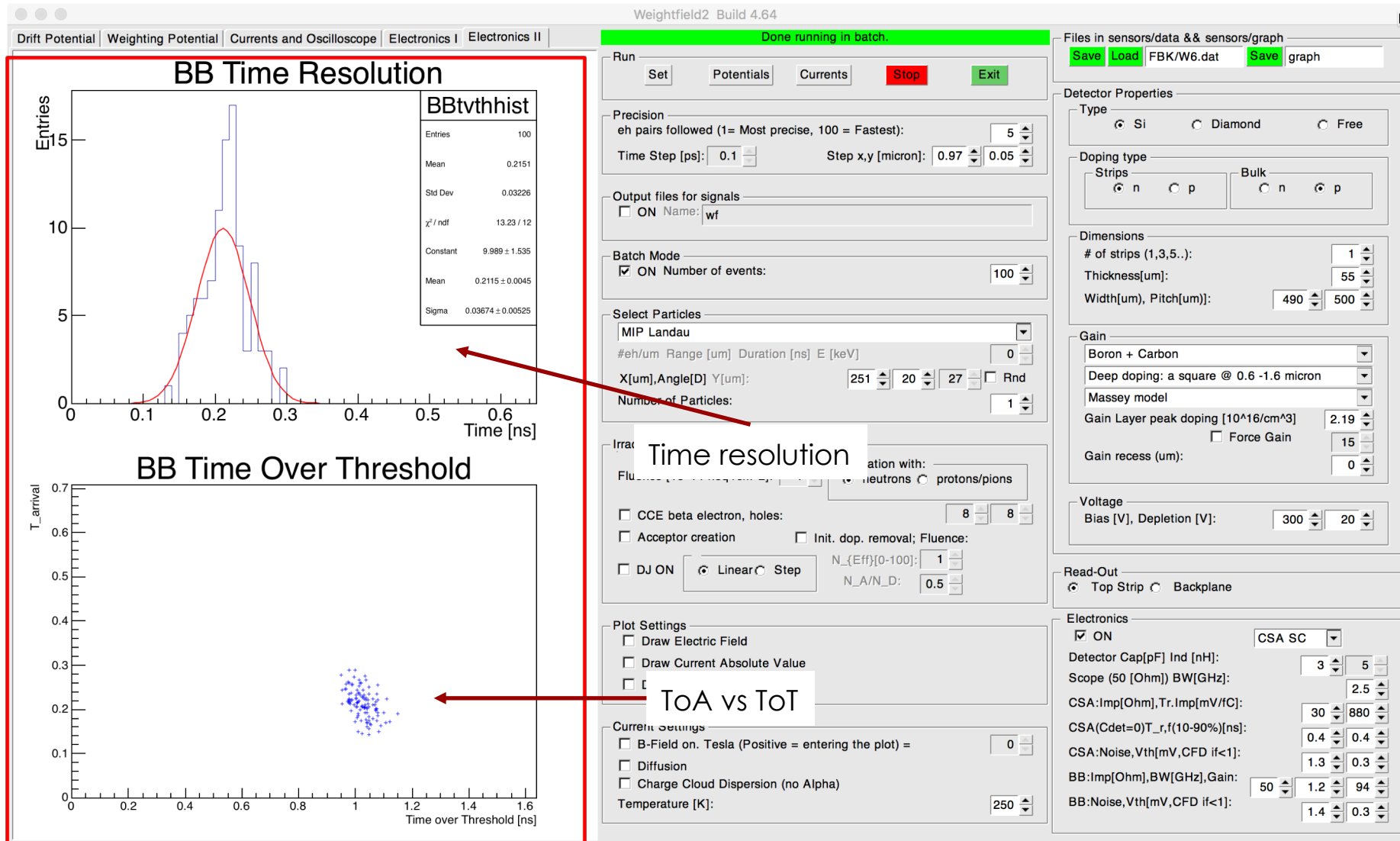


- ➔ Set the comparator threshold as low as you can
- ➔ Use thin sensors

Batch mode: deposited & collected charges

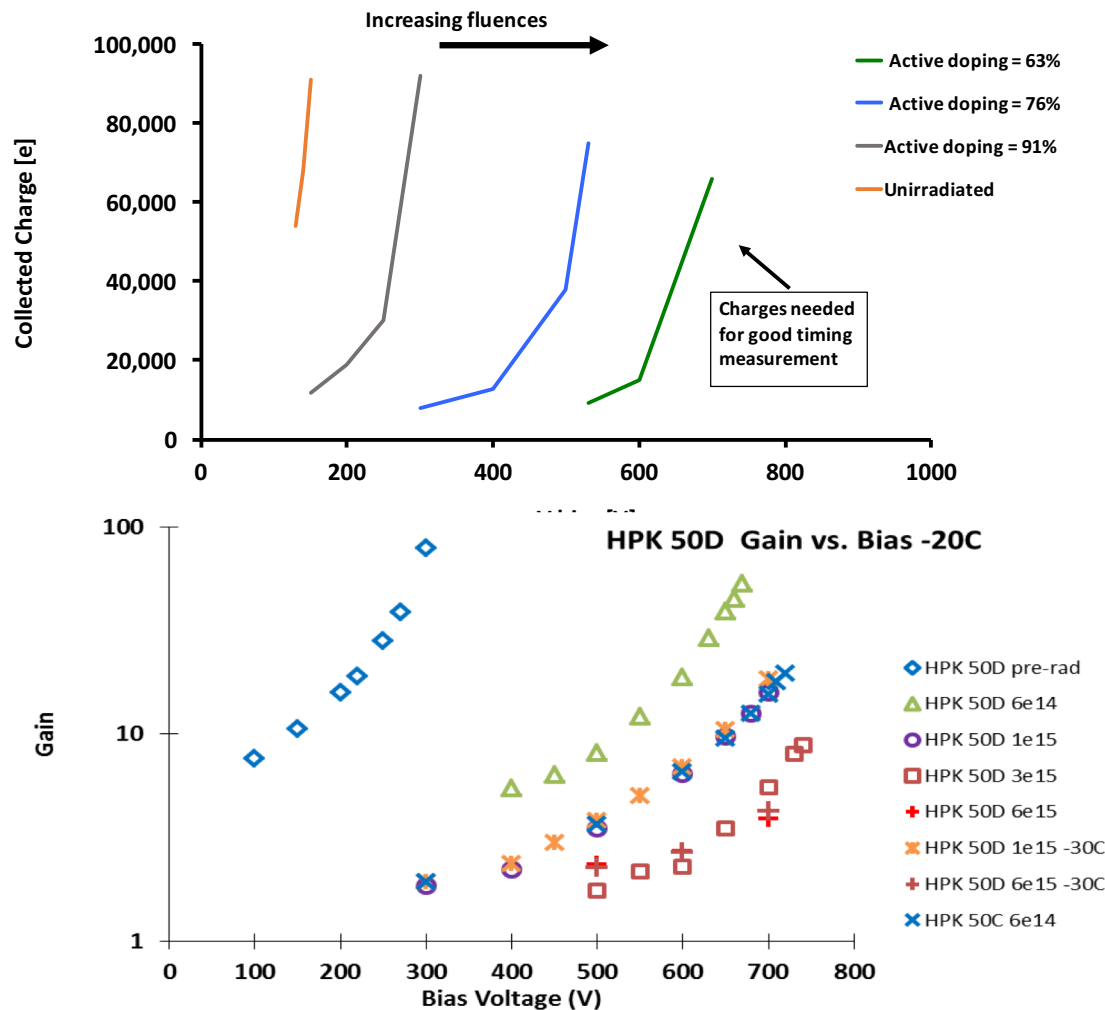


Batch mode: time resolution



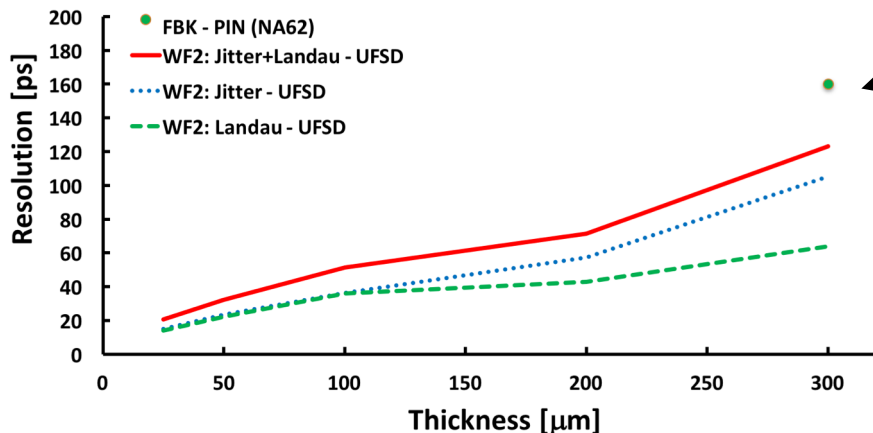
Compensation with Vbias

The necessary field can be recovered by increasing the external Vbias: proven to work up to $5 \cdot 10^{15} \text{ n}^{\text{eq}}/\text{cm}^2$

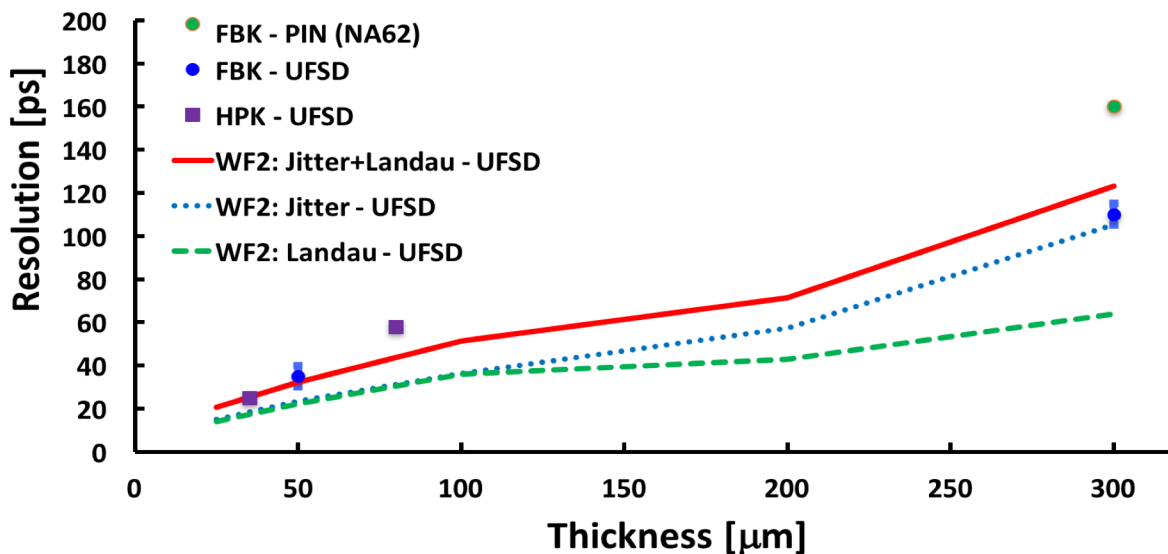


Time resolution vs thickness

Comparison WF2 Simulation - Data
Band bars show variation with temperature ($T = -20^{\circ}\text{C} - 20^{\circ}\text{C}$), and gain ($G = 20 - 30$)

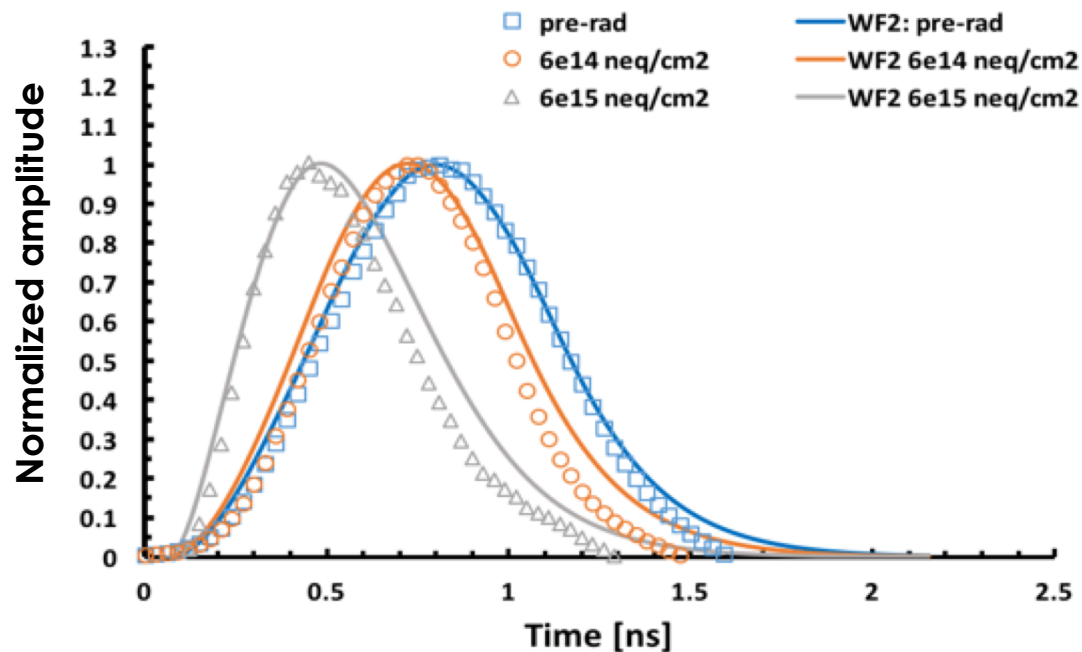


Comparison WF2 Simulation - Data
Band bars show variation with temperature ($T = -20^{\circ}\text{C} - 20^{\circ}\text{C}$), and gain ($G = 20 - 30$)



Pulse shape in irradiated UFSD

Comparison measured - WF2 pulse of HPK 50D 50-micron thick sensors

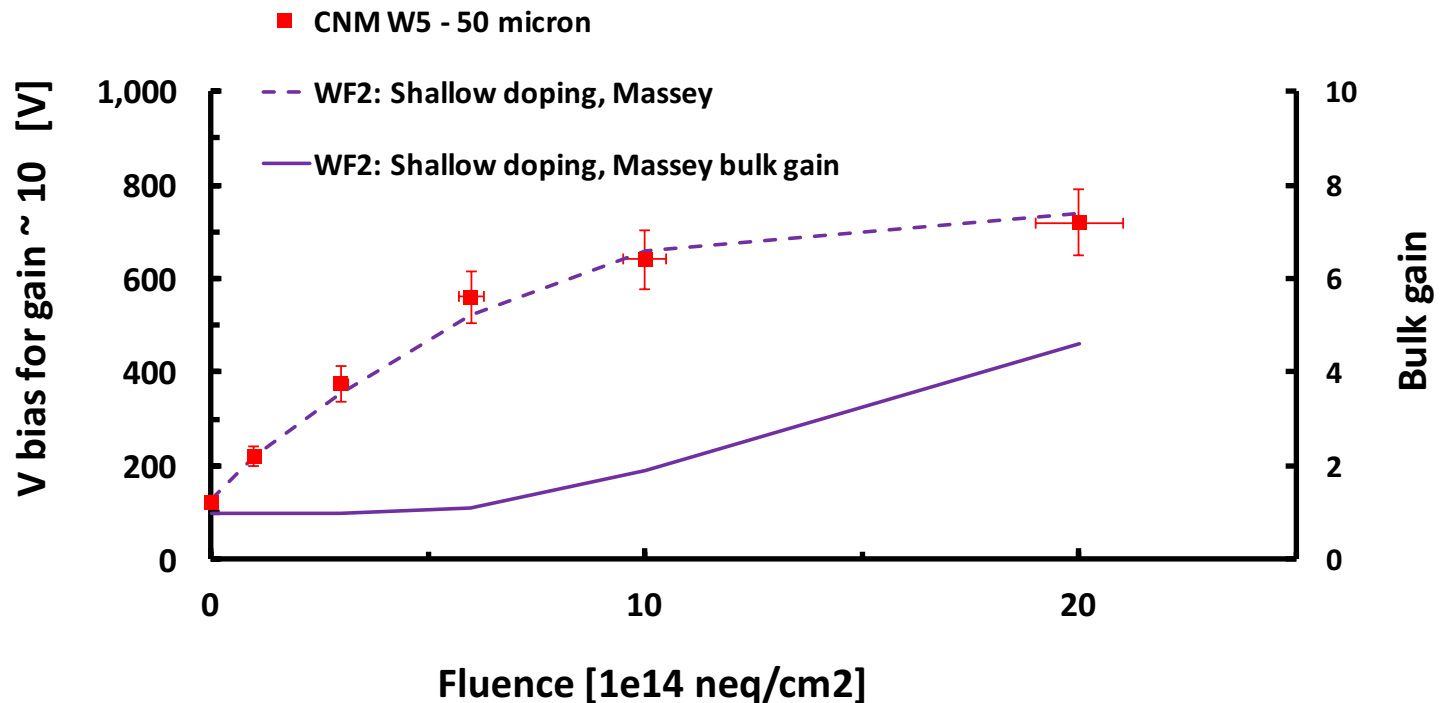


With irradiation the signal changes: it becomes shorter and steeper

How to use UFSD up to $5 \sim 10^{15} \text{ n}_{\text{eq}}/\text{cm}^2$

As the gain layer density decreases, we need to increase the external voltages to create the Efield needed for multiplications. In so doing, the gain moves from the gain layer to the bulk

Bias voltage to obtain Gain ~ 10 as a function of fluence



Conclusion

Weightfield2 is a rather easy to use simulator for silicon sensors

It can help the user's intuition in deciding the best solutions

It is fully configurable by the user

Acknowledgement

This research was carried out with the contribution of the Ministero degli Affari Esteri, "Direzione Generale per la Promozione del Sistema Paese" of Italy.



*Ministero degli Affari Esteri
e della Cooperazione Internazionale*

DIREZIONE GENERALE
PER LA PROMOZIONE DEL SISTEMA PAESE
*Unità per la cooperazione scientifica
e tecnologica bilaterale e multilaterale*

The work is supported by HORIZON2020 Grants UFSD ERC grant
UFSD669529